

Erdős Problem #629

EP-629

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

The list chromatic number $\chi_L(G)$ is defined to be the minimal k such that for any assignment of a list of k colours to each vertex of G (perhaps different lists for different vertices) a colouring of each vertex by a colour on its list can be chosen such that adjacent vertices receive distinct colours.

Determine the minimal number of vertices $n(k)$ of a bipartite graph G such that $\chi_L(G) > k$.

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>